



SOFTWARE DEV

MERN STACK DEVELOPMENT REPORT

: Application Deployment & Monetization Architecture

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1. Executive Summary

This report outlines the finalized deployment pipeline and monetization architecture for the upcoming Android application release. The strategic focus centers on establishing a compliant, scalable workflow that safely transitions the application from a local testing environment into the Google Play Store's production track. The core objective is to maximize ad revenue potential while adhering strictly to Google's updated developer compliance policies.

2. Technical Infrastructure & Monetization Strategy

The application's monetization model utilizes a gated premium utility framework, balancing user experience with high-yield revenue generation.

- **Rewarded Video Architecture:** The system employs rewarded video advertisements to temporarily unlock advanced in-app capabilities. This format was selected because rewarded video ad units consistently deliver higher Effective Cost Per Mille (eCPM) rates compared to standard display banners, optimizing the revenue generated per active user.
- **Compliance & Fraud Prevention:** To prevent account flags or invalid traffic suspensions during the QA and closed testing phases, the codebase is currently configured with official Google **Test Ad Unit IDs**.
- **Callback Logic Integration:** The Google Mobile Ads SDK event handlers are explicitly tied to the application's state management. Specifically, the `onUserEarnedReward` callback strictly dictates the feature unlock mechanism, ensuring premium capabilities are only accessible upon verified completion of the advertisement.

3. Google Play Deployment Pipeline

Publishing a new application under a personal developer account requires navigating Google Play's mandatory testing and compliance gates.

Phase	Core Objective	Key Deliverables & Requirements
1. Local	Codebase	Integrate AdMob SDK,

Phase	Core Objective	Key Deliverables & Requirements
Integration	Preparation	implement test IDs, and compile the release Android App Bundle (.aab).
2. Closed Testing	Google Compliance Gate	Distribute the build to a minimum of 12 opted-in testers for 14 consecutive days .
3. Production Release	Public Availability	Swap Test IDs with Real Ad Unit IDs , request Production Access, and publish to the Play Store.
4. Monetization	Ad Review & Payout Setup	Pass the App Readiness Review, link the store listing in AdMob, and complete payment verification.

3.1 Closed Testing Compliance Requirements

As of recent policy updates, Google Play mandates that new personal developer accounts successfully complete a rigorous closed testing phase.

- **Tester Quota & Duration:** The application must maintain a minimum of 12 active, opted-in testers for 14 continuous days.
- **Engagement Monitoring:** Google actively monitors this phase for genuine user engagement and device integrity. If the active tester count drops below 12 at any point during the testing window, the 14-day mandatory waiting period resets.

4. Financial Mechanics & Payout Logistics

Following a successful production launch and the completion of AdMob's App Readiness Review, revenue accrual and disbursement follow a standardized pipeline.

- **Real-Time Bidding (eCPM):** Advertisers dynamically bid for ad impressions based on user demographics, ad format, and geography. This establishes the eCPM, dictating the revenue generated per 1,000 ad impressions.

- **Identity & Location Verification (\$10 USD Threshold):** Upon accruing the first \$10 in finalized earnings, Google automatically dispatches a physical PIN mailer to verify the developer's registered billing address.
- **Disbursement (\$100 USD Threshold):** Once total accrued earnings meet or exceed the \$100 minimum threshold, funds are automatically disbursed via direct wire transfer during the standard monthly payout cycle (typically processed around the 21st of the subsequent month).

5. Immediate Action Items

The immediate next steps involve validating the QA environment to ensure all feature unlock callbacks function properly using the AdMob Test Ad IDs. Once verified, the initial .aab release bundle will be compiled and signed. Concurrently, the Google Play Console store listing must be finalized, including all visual assets, descriptions, and the mandatory Privacy Policy URL. Finally, the bundle will be uploaded to the Closed Testing track, and the opt-in link will be distributed to secure the 12 testers and initiate the 14-day compliance countdown.